If you are wondering how software controls the bus, you need to understand how buses are used. The bus is a set of conducting wires used by all chips connected to it. When a device wants to send data, it sends the electric signals via these wires. This is very simple till you question "what happens when two devices want to send the data to the processor at the same time?" If two or more components were to start sending data at the same time over the bus, all data would reach corrupted because the electric signals from multiple devices would cause interference with each other.

For this, a system of interconnection is built which governs how devices can send data over bus at the same time (different modules using different interconnections), or one after another (scheduling the usage of bus) depending on how the bus was architected. Bus design and implementation is a vast and interesting field.

In case you're wondering "why C?", the reason is "performance". The C programming language compiles to binary code and offers capabilities to control the hardware easily. Most operating system kernels are written in C and so are most drivers. It is easier to write code in C than in any platform's assembly language while intelligent compilers ensure that the generated binary remains performant.

How do Apps run?

Software we have been talking about till now are system-level programs. Though apps are not the focus of this chapter, it's time we talk about the wheels of the smartphone revolution – apps!

The word 'application' means to put something in operation. Application programs are software designed to put the hardware into operation for a task. We have already discussed that operating systems are designed to make sure that certain actions are privileged and to perform those operations, the OS exposes APIs (Application Programming Interfaces).

Applications are normally compiled for a certain OS and architecture so that it can use the capabilities of the target OS and run code on the target microprocessor. When the application is launched, the OS reads the application file, loads executable code in memory and asks the processor to start executing the loaded executable with the lowest privilege levels. As the application executes, it calls APIs provided by the OS for its needs (like reading file from disk, sending data over network etc.).